Motion in a plane CHEAT SHEET

VECTORS: ADDING AND THE RESULTANT

scalar: a number + unit (2 kg) · vector: size + direction

12 km E then 5 km N: flown 17 km, displacement 13 km at 22.6°

add head to tail: the resultant runs first tail → last head

$A + B = B + A$ (triangle = parallelogram law)

$R^2 = A^2 + B^2 + 2AB\cos\theta$ (θ tail to tail)

$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$ 4 km, 3 km, 60° → 6.08 km, 25.3°

edges: $\theta = 0$ → $A + B$ (7) · 90° → Pythagoras (5) · 180° → $A - B$ (1)

$A - B = A + (-B)$: Δv for 10 m/s E → 10 m/s N = 14.14 m/s NW

UNIT VECTORS AND COMPONENTS

unit vector $\hat{A} = A/|A|$; $3\hat{i} + 4\hat{j} \rightarrow 0.6\hat{i} + 0.8\hat{j}$

$A_x = A\cos\theta$, $A_y = A\sin\theta$ (θ from the +x axis)

20 m/s at 30° → 17.32 across, 10.00 up

$|A| = \sqrt{(A_x^2 + A_y^2)}$, $\tan\theta = A_y/A_x$ (sketch the quadrant)

(−6, 8) → 10 m/s at 126.87°, not −53.13°

add across to across, up to up: (4, 0) + (1.5, 2.6) → 6.08 km

Python: `20 * math.cos(math.radians(30)) = 17.32` (radians)

THE DOT AND CROSS PRODUCTS

$A \cdot B = AB\cos\theta = A_xB_x + A_yB_y$ (a number)

the part of B along A; $\theta = 90° \rightarrow 0$; $\theta > 90°$ → negative

$(2, 1)\cdot(1, 3) = 5 \rightarrow \cos\theta = 5/\sqrt{50} \rightarrow \theta = 45°$

$|A \times B| = AB\sin\theta$, perpendicular to both (right-hand rule); $\hat{i} \times \hat{j} = \hat{k}$

$A \times B = (A_xB_y - A_yB_x)\,\hat{k} = -B \times A$; area of the parallelogram

$3\hat{i}$ and $2\hat{i} + 2\hat{j} \rightarrow 6\hat{k}$; parallel → 0

MOTION IN A PLANE

$v = \frac{dr}{dt}$ ($v_x = dx/dt$: the slope of x–t), along the path's tangent

average velocity $\Delta r/\Delta t$: (15, 9) m in 3 s → (5, 3) m/s

$a = \frac{dv}{dt}$; turning at a steady speed is still accelerating

constant a, each axis alone: $v = u + at$

$r = ut + \frac{1}{2}at^2$ (the area under v–t: rectangle + triangle)

$u = 5\hat{i}$, $a = 2\hat{j}$, t = 3 s → $v = 5\hat{i} + 6\hat{j}$, $r = 15\hat{i} + 9\hat{j}$

edges: $a = 0$ → a straight line; u parallel to a → still a line

Motion in a plane CHEAT SHEET

PROJECTILES

$x = u\cos\theta \cdot t$, $y = u\sin\theta \cdot t - \frac{1}{2}gt^2$ ($g = 9.8$ m/s², no air)

$v_x = u\cos\theta$ (constant), $v_y = u\sin\theta - gt$

time to the top $\frac{u\sin\theta}{g}$, time of flight $T = \frac{2u\sin\theta}{g}$

maximum height $H = \frac{u^2\sin^2\theta}{2g}$

20 m/s at 30°: $T = 2.04$ s, $H = 5.10$ m

level launch from h: $t = \sqrt{(2h/g)}$; 1.25 m at 2 m/s → 0.505 s, 1.01 m

path $y = x\tan\theta - \frac{g\,x^2}{2u^2\cos^2\theta}$ (a parabola)

20 m/s, 30°, at $x = 30$ m: $y = 17.32 - 14.70 = 2.62$ m

THE RANGE

range $R = \frac{u^2\sin 2\theta}{g}$ (level ground)

20 m/s at 30° → 35.35 m (simulated: within 1 mm)

45° reaches farthest: $u^2/g = 40.82$ m at 20 m/s

θ and $90° - \theta$ land together: 30° and 60° → 35.35 m

the sprinkler, 15 m/s: 45° → 22.96 m; 30° or 60° → 19.88 m

backwards: $u = \sqrt{(Rg/\sin 2\theta)}$: 45 m at 35° → 21.66 m/s

$\sin 2\theta = Rg/u^2$: 18 m at 15 m/s → 25.8° or 64.2°

edges: 0° and 90° → 0; Moon ($g \approx 1.62$) → ≈ 6 times farther

Python: math.sin(math.radians(2 * θ)); without it: −12.44 m

UNIFORM CIRCULAR MOTION

period T: one turn; 1 rad: arc = radius; $s = r\theta$

$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T}$, $v = \omega r$ (the farther out, the faster)

r 25 m, v 10 m/s: $T = 15.71$ s, $\omega = 0.4$ rad/s

centripetal $a = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2}$, towards the centre

the car: $10^2/25 = 4$ m/s² (simulated: stays on the circle)

fan tip 0.6 m, 4 turns/s: 15.08 m/s, 379 m/s² (≈ 39 g)

backwards: $v = \sqrt{(ar)}$: 6 m/s² on 50 m → 17.32 m/s

edges: $r \to \infty \to a \to 0$, a straight line; $r = 0 \to v = 0$

CHECK BEFORE YOU TRUST IT

before you trust an answer:

1 did I split every vector into across and up parts?

2 does each axis use only its own numbers?

3 does the formula's condition hold (level ground, no air)?

4 does a sketch agree (quadrant, which root, which way)?